University of British Columbia, Vancouver
Department of Computer Science

CPSC 304 Project Cover Page

Milestone #: 2

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Group Number: 54

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

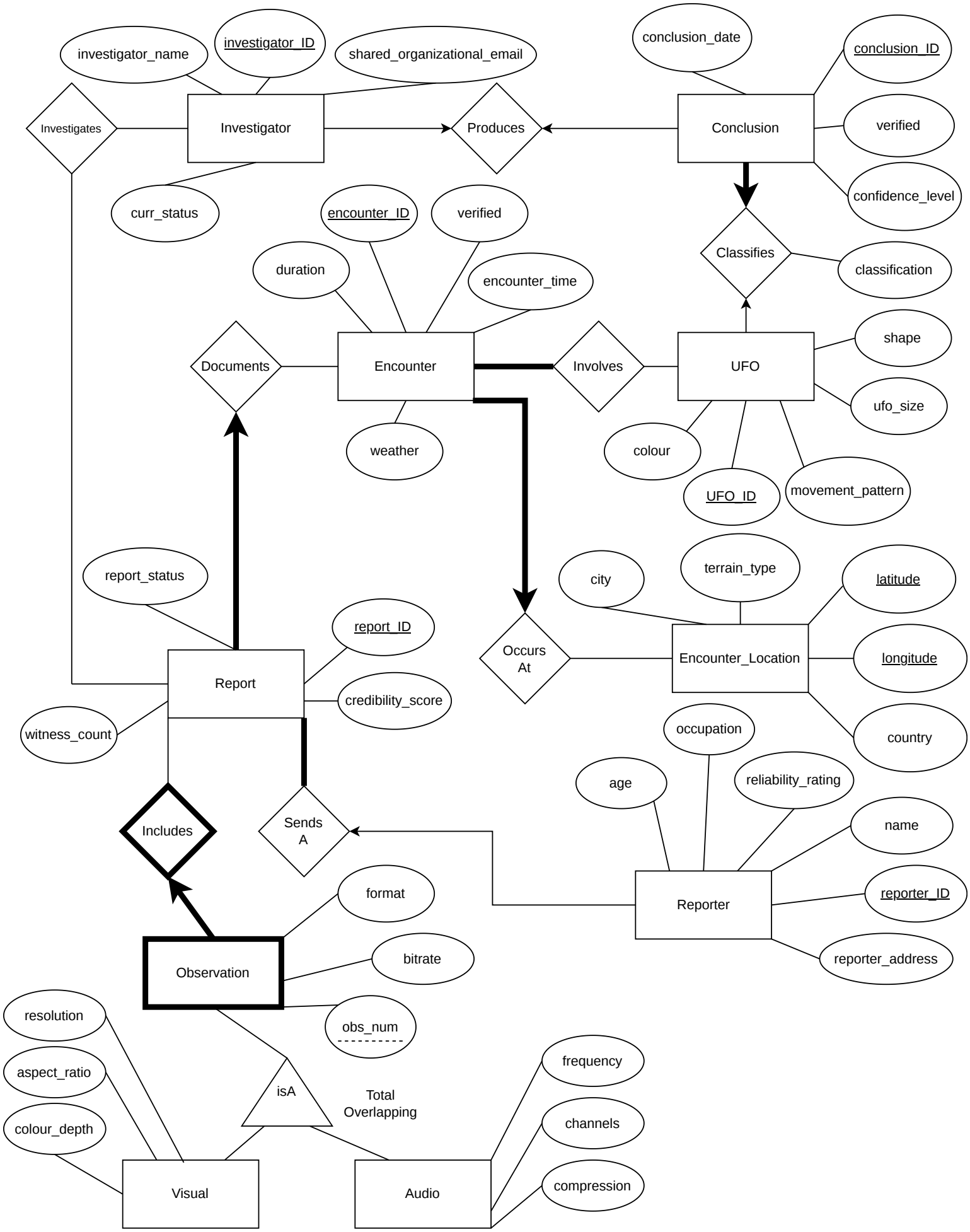
In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Summary:

Project is a database application that models UFO sightings that includes entities such as: reports, reporter, observations, encounter details, location, investigator, conclusion, and the UFO involved. This provides a centralized and easy to access source for UFO sightings to be reported and investigated.

Database/Programming Language:

Database: Department provided Oracle database
Programming Language: JavaScript/React


investigator_name
investigator_ID
shared_organizational_email
conclusion_date
conclusion_ID
Investigates
Investigator
Produces
Conclusion
verified
confidence_level
curr_status
encounter_ID
verified
Classifies
classification
duration
encounter_time
Documents
Encounter
Involves
UFO
shape
ufo_size
weather
colour
UFO_ID
movement_pattern
report_status
city
terrain_type
latitude
report_ID
Occurs At
Encounter_Location
longitude
Report
credibility_score
witness_count
country
occupation
age
reliability_rating
Includes
Sends A
name
format
Reporter
reporter_ID
Observation
bitrate
reporter_address
resolution
obs_num
aspect_ratio
isA
Total Overlapping
frequency
colour_depth
channels
Visual
Audio
compression

Derived Schema:

Investigator(investigator_ID: integer, investigator_name: char[100], shared_organizational_email: char[150], curr_status: char[30])

- PK: investigator_ID
- CK: none
- FK: none
- Other constraints: investigator_name NOT NULL, curr_status NOT NULL

Investigates(**investigator_ID:** integer, **report_ID**: integer)

- PK: investigator_ID, report_ID
- CK: none
- FK: investigator_ID -> Investigator(investigator_ID), report_ID -> Report(report_ID)
- Other constraints: none

Conclusion(conclusion_ID: integer, **investigator_ID**: integer, conclusion_date: time, verified: char(1), confidence_level: integer)

- PK: conclusion_ID
- CK: investigator_ID (investigator_ID is UNIQUE)
- FK: investigator_ID -> Investigator(investigator_ID)
- Other constraints: investigator_ID UNIQUE and NOT NULL

Classifies(**conclusion_ID**: integer, **UFO_ID**: integer, classification: char(50))

- PK: conclusion_ID, UFO_ID
- FK: conclusion_ID -> Conclusion(conclusion_ID), UFO_ID -> UFO(UFO_ID)
- CK: conclusion_ID; UFO_ID (both UNIQUE)
- Other constraints: UNIQUE (conclusion_ID), UNIQUE (UFO_ID)

Encounter(encounter_ID: integer, **latitude**: decimal(20, 17)**, longitude**: decimal(20, 17), duration: integer, verified: char(1), encounter_time: time, weather: char(50))

- PK: encounter_ID
- CK: N/A
- FK: (latitude, longitude) -> Encounter_Location(latitude, longitude)
- Other constraints: duration > 0, latitude NOT NULL, longitude NOT NULL

UFO(UFO_ID: integer, ufo_size: integer, colour: char(30), shape: char(30), movement_pattern: char(50))

- PK: UFO_ID
- CK: N/A
- FK: none
- Other constraints: ufo_size> 0

Involves(**encounter_ID**: integer, **UFO_ID**: integer)

- PK: encounter_ID, UFO_ID
- CK: N/A
- FK: encounter_ID -> Encounter(encounter_ID), UFO_ID -> UFO(UFO_ID)
- Other constraints: ON DELETE CASCADE

Report(report_ID: integer, **encounter_ID**: integer, **reporter_ID**: integer, witness_count: integer, report_status: char(40), credibility_score: integer)

- PK: report_ID
- CK: N/A
- FK: encounter_ID -> Encounter(encounter_ID), reporter_ID -> Reporter(reporter_ID)
- Other constraints: witness_count > 0, encounter_ID NOT NULL, reporter_ID NOT NULL

Encounter_Location(latitude: decimal(20, 17), longitude: decimal(20, 17), city: char(100), terrain_type: char(50), country: char(60))

- PK: latitude, longitude
- CK: N/A
- FK: N/A
- Other constraints: none

Reporter(reporter_ID: integer, occupation: char(60), reliability_rating: integer, name: char(100), age: integer, reporter_address: char(200))

- PK: reporter_ID
- CK: N/A
- FK: N/A
- Other constraints: none

Observation(**report_ID**: integer, obs_num: integer, bitrate: integer, format: char(20))

- PK: report_ID, obs_num
- CK: N/A
- FK: report_ID -> Report(report_ID)
- Other constraints: none

Visual(**report_ID**: integer, **obs_num**: integer, resolution: char(30), aspect_ratio: char(20), colour_depth: integer)

- PK: report_ID, obs_num
- CK: N/A
- FK: (report_ID, obs_num) -> Observation(report_ID, obs_num)
- Other constraints: none

Audio(**report_ID**: integer, **obs_num**: integer, frequency: integer, channels: integer, compression: integer)

- PK: report_ID, obs_num
- CK: N/A
- FK: (report_ID, obs_num) -> Observation(report_ID, obs_num)
- Other constraints: none

Functional Dependencies:

Investigator(investigator_ID, investigator_name, shared_organizational_email, curr_status)

- investigator_ID -> investigator_name, shared_organizational_email, curr_status

Investigates(**investigator_ID**, **report_ID**)

- No non-trivial FDs

Conclusion(conclusion_ID, **investigator_ID**, conclusion_date, verified, confidence_level)

- conclusion_ID -> investigator_ID, conclusion_date, verified, confidence_level
- investigator_ID -> conclusion_ID, conclusion_date, verified, confidence_level

Encounter(encounter_ID, **latitude, longitude**, duration, verified, encounter_time, weather)

- encounter_ID -> latitude, longitude, duration, verified, encounter_time, weather

UFO(UFO_ID, ufo_size, colour, shape, movement_pattern)

- UFO_ID -> ufo_size, colour, shape, movement_pattern

Involves(**encounter_ID**, **UFO_ID**)

- No non-trivial FDs

Report(report_ID, **encounter_ID**, **reporter_ID**, witness_count, report_status, credibility_score)

- report_ID -> encounter_ID, reporter_ID, witness_count, report_status, credibility_score

Encounter_Location(latitude, longitude, city, terrain_type, country)

- Latitude, longitude -> city, terrain_type, country

Observation(**report_ID**, obs_num, bitrate, format)

- report_ID, obs_num -> bitrate, format

Reporter(reporter_ID, occupation, reliability_rating, name, age, reporter_address)

- reporter_ID -> occupation, reliability_rating, name, age, reporter_address

Visual(**report_ID**, **obs_num**, resolution, aspect_ratio, colour_depth)

- report_ID, obs_num -> resolution, colour_depth
- resolution -> aspect_ratio

Audio(**report_ID**, **obs_num**, frequency, channels, compression)

- report_ID, obs_num -> frequency, channels, compression

Classifies(**conclusion_ID**, **UFO_ID**, classification)

- conclusion_ID, UFO_ID -> classification
- conclusion_ID -> UFO_ID, classification
- UFO_ID -> conclusion_ID, classification

Normalization

Part I: Normalization Process

Investigator(investigator_ID, investigator_name, shared_organizational_email, curr_status)

FD: investigator_ID -> investigator_name, shared_organizational_email, curr_status
- investigator_ID is a superkey -> BCNF

Investigates(**investigator_ID**, **report_ID**)

- Has no non key attributes
- (investigator_ID, report_ID) is a superkey -> BCNF

Conclusion(conclusion_ID, **investigator_ID**, conclusion_date, verified, confidence_level)

- conclusion_ID -> investigator_ID, conclusion_date, verified, confidence_level
- investigator_ID -> conclusion_ID, conclusion_date, verified, confidence_level
- Every LHS of a FD is a key -> BCNF

Encounter(encounter_ID, **latitude, longitude**, duration, verified, encounter_time, weather)

- encounter_ID -> latitude, longitude, duration, verified, encounter_time, weather
- Every LHS of a FD is a key -> BCNF

UFO(UFO_ID, ufo_size, colour, shape, movement_pattern)

- UFO_ID -> ufo_size, colour, shape, movement_pattern
- Every LHS of a FD is a key -> BCNF

Involves(**encounter_ID**, **UFO_ID**)

- Has no non key attributes
- (encounter_ID, UFO_ID) is a superkey -> BCNF

Report(report_ID, **encounter_ID**, **reporter_ID**, witness_count, report_status, credibility_score)

- report_ID -> encounter_ID, reporter_ID, witness_count, report_status, credibility_score
- Every LHS of a FD is a key -> BCNF

Encounter_Location(latitude, longitude, city, terrain_type, country)

- Latitude, longitude -> city, terrain_type, country
- Every LHS of a FD is a key -> BCNF

Observation(**report_ID**, obs_num, bitrate, format)

- report_ID, obs_num -> bitrate, format
- Every LHS of a FD is a key -> BCNF

Reporter(reporter_ID, occupation, reliability_rating, name, age, reporter_address)

- reporter_ID -> occupation, reliability_rating, name, age, reporter_address
- Every LHS of a FD is a key -> BCNF

(pre-normalized) Visual(**report_ID**, **obs_num**, resolution, aspect_ratio, colour_depth)

- report_ID, obs_num -> resolution, aspect_ratio, colour_depth
- resolution -> aspect_ratio

Since resolution is not a superkey, resolution -> aspect_ratio violates BCNF
Decompose on resolution -> aspect_ratio

R(report_ID, obs_num, resolution, aspect_ratio, colour_depth)

R1 = (resolution, aspect_ratio)
R2 = (report_ID, obs_num, resolution, colour_depth)

- R1: resolution is key in resolution -> aspect_ratio -> BCNF
- R2: (report_ID, obs_num) is the key in (report_ID, obs_num) -> resolution, colour_depth -> BCNF
- **Relations after normalization**:
    1. ResolutionAspect(**resolution**: char(30), aspect_ratio: char(20)),
    2. Visual(**report_ID**: integer, **obs_num**: integer, resolution: char(30), colour_depth: integer)

Audio(**report_ID**, **obs_num**, frequency, channels, compression)

- report_ID, obs_num -> frequency, channels, compression

- Every LHS of a FD is a key -> BCNF

Classifies(**conclusion_ID**, **UFO_ID**, classification)

- conclusion_ID, UFO_ID -> classification
- conclusion_ID -> UFO_ID, classification
- UFO_ID -> conclusion_ID, classification
- Every LHS of a FD is a key -> BCNF

Part 2: Tables

Investigator(investigator_ID: integer, investigator_name: char[100], shared_organizational_email: char[150], curr_status: char[30])

- PK: investigator_ID
- CK: none
- FK: none
- Other constraints: investigator_name NOT NULL, curr_status NOT NULL

Investigates(**investigator_ID:** integer, **report_ID**: integer)

- PK: investigator_ID, report_ID
- CK: none
- FK: investigator_ID -> Investigator(investigator_ID), report_ID -> Report(report_ID)
- Other constraints: none

Conclusion(conclusion_ID: integer, **investigator_ID**: integer, conclusion_date: time, verified: char(1), confidence_level: integer)

- PK: conclusion_ID
- CK: investigator_ID (investigator_ID is UNIQUE)
- FK: investigator_ID -> Investigator(investigator_ID)
- Other constraints: investigator_ID UNIQUE and NOT NULL

Classifies(**conclusion_ID**: integer, **UFO_ID**: integer, classification: char(50))

- PK: conclusion_ID, UFO_ID
- FK: conclusion_ID -> Conclusion(conclusion_ID), UFO_ID -> UFO(UFO_ID)
- CK: conclusion_ID; UFO_ID (both UNIQUE)
- Other constraints: UNIQUE (conclusion_ID), UNIQUE (UFO_ID)

Encounter(encounter_ID: integer, **latitude**: decimal(20, 17)**, longitude**: decimal(20, 17), duration: integer, verified: char(1), encounter_time: time, weather: char(50))

- PK: encounter_ID
- CK: N/A
- FK: (latitude, longitude) -> Encounter_Location(latitude, longitude)
- Other constraints: duration > 0, latitude NOT NULL, longitude NOT NULL

UFO(<u>UFO_ID</u>: integer, ufo_size: integer, colour: char(30), shape: char(30), movement_pattern: char(50))

- PK: UFO_ID
- CK: N/A
- FK: none
- Other constraints: ufo_size> 0

Involves(**<u>encounter_ID</u>**: integer, **<u>UFO_ID</u>**: integer)

- PK: encounter_ID, UFO_ID
- CK: N/A
- FK: encounter_ID -> Encounter(encounter_ID), UFO_ID -> UFO(UFO_ID)
- Other constraints: ON DELETE CASCADE

Report(<u>report_ID</u>: integer, **encounter_ID**: integer, **reporter_ID**: integer, witness_count: integer, report_status: char(40), credibility_score: integer)

- PK: report_ID
- CK: N/A
- FK: encounter_ID -> Encounter(encounter_ID), reporter_ID -> Reporter(reporter_ID)
- Other constraints: witness_count > 0, encounter_ID NOT NULL, reporter_ID NOT NULL

Encounter_Location(<u>latitude</u>: decimal(20, 17), <u>longitude</u>: decimal(20, 17), city: char(100), terrain_type: char(50), country: char(60))

- PK: latitude, longitude
- CK: N/A
- FK: N/A
- Other constraints: none

Reporter(<u>reporter_ID</u>: integer, occupation: char(60), reliability_rating: integer, name: char(100), age: integer, reporter_address: char(200))

- PK: reporter_ID
- CK: N/A
- FK: N/A
- Other constraints: none

Observation(**<u>report_ID</u>**: integer, <u>obs_num</u>: integer, bitrate: integer, format: char(20))

- PK: report_ID, obs_num
- CK: N/A
- FK: report_ID -> Report(report_ID)
- Other constraints: none

Visual(**<u>report_ID</u>**: integer, **<u>obs_num</u>**: integer, resolution: char(30), colour_depth: integer)

- PK: report_ID, obs_num

- CK: N/A
- FK: (report_ID, obs_num) -> Observation(report_ID, obs_num), resolution -> ResolutionAspect(resolution)
- Other constraints: none

ResolutionAspect(**<u>resolution</u>:** char(30), aspect_ratio: char(20))

- PK: resolution
- CK: N/A
- FK: N/A
- Other constraints: none

Audio(**<u>report_ID</u>**: integer, **<u>obs_num</u>**: integer, frequency: integer, channels: integer, compression: integer)

- PK: report_ID, obs_num
- CK: N/A
- FK: (report_ID, obs_num) -> Observation(report_ID, obs_num)
- Other constraints: none

SQL DDL

```
-- Core entity tables

CREATE TABLE UFO(
   UFO_ID INTEGER,
   ufo_size INTEGER,
   colour VARCHAR2(30),
   shape VARCHAR2(30),
   movement_pattern VARCHAR2(50),
   PRIMARY KEY (UFO_ID)
);

CREATE TABLE Investigator(
   investigator_ID INTEGER,
   investigator_name VARCHAR2(100) NOT NULL,
   shared_organizational_email VARCHAR2(150),
   curr_status VARCHAR2(30) NOT NULL,
   PRIMARY KEY (investigator_ID)
);

CREATE TABLE Encounter_Location(
   latitude NUMBER(20, 17),
   longitude NUMBER(20, 17),
   city VARCHAR2(100),
   terrain_type VARCHAR2(50),
   country VARCHAR2(60),
   PRIMARY KEY (latitude, longitude)
```

```
);

CREATE TABLE Encounter(
   encounter_ID INTEGER,
   latitude NUMBER(20, 17),
   longitude NUMBER(20, 17),
   duration INTEGER,
   verified CHAR(1),
   encounter_time TIMESTAMP,
   weather VARCHAR2(50),
   PRIMARY KEY (encounter_ID),
   FOREIGN KEY (latitude, longitude) REFERENCES Encounter_Location(latitude, longitude)
);

CREATE TABLE Reporter(
   reporter_ID INTEGER,
   occupation VARCHAR2(60),
   reliability_rating INTEGER,
   reporter_name VARCHAR2(100),
   age INTEGER,
   reporter_address VARCHAR2(200),
   PRIMARY KEY (reporter_ID)
);

CREATE TABLE Report(
   report_ID INTEGER,
   encounter_ID INTEGER NOT NULL,
   reporter_ID INTEGER NOT NULL,
   witness_count INTEGER,
   report_status CHAR(40),
   credibility_score INTEGER,
   PRIMARY KEY (report_ID),
   FOREIGN KEY (encounter_ID) REFERENCES Encounter(encounter_ID),
   FOREIGN KEY (reporter_ID) REFERENCES Reporter(reporter_ID)
);

CREATE TABLE Conclusion(
   conclusion_ID INTEGER,
   investigator_ID INTEGER NOT NULL UNIQUE,
   conclusion_date TIMESTAMP,
   verified CHAR(1),
   confidence_level INTEGER,
   PRIMARY KEY (conclusion_ID),
   FOREIGN KEY (investigator_ID) REFERENCES Investigator(investigator_ID)
);

-- Relationship tables

CREATE TABLE Investigates(
   investigator_ID INTEGER NOT NULL,
   report_ID INTEGER NOT NULL,
```

```
   PRIMARY KEY (investigator_ID, report_ID),
   FOREIGN KEY (investigator_ID) REFERENCES Investigator(investigator_ID) ON DELETE
CASCADE,
   FOREIGN KEY (report_ID) REFERENCES Report(report_ID) ON DELETE CASCADE
);

CREATE TABLE Involves(
   encounter_ID INTEGER,
   UFO_ID INTEGER,
   PRIMARY KEY (encounter_ID, UFO_ID),
   FOREIGN KEY (encounter_ID) REFERENCES Encounter(encounter_ID) ON DELETE
CASCADE,
   FOREIGN KEY (UFO_ID) REFERENCES UFO(UFO_ID) ON DELETE CASCADE
);

CREATE TABLE Classifies(
   conclusion_ID INTEGER NOT NULL,
   UFO_ID INTEGER NOT NULL,
   classification VARCHAR2(50),
   PRIMARY KEY (conclusion_ID, UFO_ID),
   UNIQUE (conclusion_ID),
   UNIQUE (UFO_ID),
   FOREIGN KEY (conclusion_ID) REFERENCES Conclusion(conclusion_ID) ON DELETE
CASCADE,
   FOREIGN KEY (UFO_ID) REFERENCES UFO(UFO_ID) ON DELETE CASCADE
);

-- Weak entity and subtypes

CREATE TABLE Observation(
   report_ID INTEGER,
   obs_num INTEGER,
   bitrate INTEGER,
   format VARCHAR2(20),
   PRIMARY KEY (report_ID, obs_num),
   FOREIGN KEY (report_ID) REFERENCES Report(report_ID) ON DELETE CASCADE
);

CREATE TABLE ResolutionAspect(
   resolution VARCHAR2(30),
   aspect_ratio VARCHAR2(20),
   PRIMARY KEY (resolution)
);

CREATE TABLE Visual(
   report_ID INTEGER,
   obs_num INTEGER,
   resolution VARCHAR2(30),
   colour_depth INTEGER,
   PRIMARY KEY (report_ID, obs_num),
```

```
   FOREIGN KEY (report_ID, obs_num) REFERENCES Observation(report_ID, obs_num) ON
DELETE CASCADE,
   FOREIGN KEY (resolution) REFERENCES ResolutionAspect(resolution)
);

CREATE TABLE Audio(
   report_ID INTEGER,
   obs_num INTEGER,
   frequency INTEGER,
   channels INTEGER,
   audio_compression VARCHAR2(20),
   PRIMARY KEY (report_ID, obs_num),
   FOREIGN KEY (report_ID, obs_num) REFERENCES Observation(report_ID, obs_num) ON
DELETE CASCADE
);
```

SQL DDL Foreign Key Handling

ON UPDATE: Primary Keys are treated as immutable identifiers and will not be updated (Oracle doesn't support ON UPDATE CASCADE).

ON DELETE: For dependent or relationship tables (Observation, Visual, Audio, Investigates, Involves, Classifies), ON DELETE CASCADE is used to prevent leftover rows. For core entity relationships (Report->Encounter, Report->Reporter, Conclusion->Investigator), cascade is not used because these are main entities in the database that should not be automatically deleted. For shared reference tables (Encounter->Encounter_Location and Visual->ResolutionAspect), cascade is not used as these rows may still be referenced by multiple tuples.

AI Tool Acknowledgement

AI tools were not used in the making of Milestone 2.